Maclaurin Series & Convergence of Series

ANSWER KEY

Section 1: The Maclaurin Series Formula

1 What is the general formula for the Maclaurin series expansion of a function $f(x)$?

a. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \dots$ ✓

b. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(0)x^n$

c. $f(x) = \sum_{n=1}^{\infty} \frac{f(n)}{n!}x^n$

d. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!}x^n$

A Maclaurin series is a Taylor series centered at $x = 0$: $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$.

2 Find the first three non-zero terms of the Maclaurin expansion of $f(x) = \cos(2x)$.

a. $1 - 2x^2 + \frac{2}{3}x^4$ ✓

b. $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

c. $1 - 4x^2 + 16x^4$

d. $2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

Standard series $\cos u = 1 - \frac{u^2}{2} + \frac{u^4}{24} - \dots$. Substitute $u = 2x$:
$1 - \frac{(2x)^2}{2} + \frac{(2x)^4}{24} = 1 - 2x^2 + \frac{16x^4}{24} = 1 - 2x^2 + \frac{2}{3}x^4$.

3 What is the coefficient of x^3 in the Maclaurin series of e^{-3x}?

a. $-\frac{9}{2}$ ✓

b. $\frac{9}{2}$

c. -9

d. -27

$e^u = \sum \frac{u^n}{n!}$. For $n = 3$, $\frac{(-3x)^3}{3!} = \frac{-27x^3}{6} = -\frac{9}{2}x^3$. The coefficient is $-\frac{9}{2}$.

Section 2: Manipulating Standard Maclaurin Expansions

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4 Find the Maclaurin series for $\ln(1+3x)$ up to the x^3 term.

a. $3x - \frac{9}{2}x^2 + 9x^3$ ✓

b. $3x + \frac{9}{2}x^2 + 9x^3$

c. $x - \frac{1}{2}x^2 + \frac{1}{3}x^3$

d. $3x - 9x^2 + 27x^3$

Standard series $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \dots$. With $u = 3x$: $(3x) - \frac{(3x)^2}{2} + \frac{(3x)^3}{3} = 3x - \frac{9}{2}x^2 + 9x^3$.

5 Evaluate $\lim_{x \to 0} \frac{e^x - 1 - x}{x^2}$ using Maclaurin series.

a. $\frac{1}{2}$ ✓

b. 1

c. 0

d. ∞

$e^x = 1 + x + \frac{x^2}{2} + \dots$. Then $e^x - 1 - x = \frac{x^2}{2} + O(x^3)$. Dividing by x^2: $\frac{1}{2} + O(x) \to \frac{1}{2}$ as $x \to 0$.

6 Find the first two non-zero terms of the series for $\frac{1}{1+x^2}$.

a. $1 - x^2$ ✓

b. $1 + x^2$

c. $1 - x$

d. $x - x^3$

Geometric series $\frac{1}{1-u} = 1 + u + u^2 + \dots$. With $u = -x^2$: $1 - x^2 + x^4 - \dots$.

Section 3: The Ratio Test for Convergence

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7 What does the Ratio Test state for a positive series $\sum a_n$ when $L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$?

a. **Converges if $L < 1$, diverges if $L > 1$, test is inconclusive if $L = 1$** ✓

b. Converges if $L = 1$, diverges otherwise

c. Diverges if $L < 1$, converges if $L > 1$

d. Always inconclusive for power series

The Ratio Test establishes absolute convergence when the limiting ratio $L < 1$, divergence when $L > 1$, and provides no information when $L = 1$.

8 Determine the convergence of $\sum_{n=1}^{\infty} \frac{n}{3^n}$ using the Ratio Test.

a. **Converges, because $L = \lim \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} = \frac{1}{3} < 1$** ✓

b. Diverges, because $L = 3 > 1$

c. Inconclusive, because $L = 1$

d. Diverges, because the terms do not approach 0

$\lim_{n\to\infty} \frac{n+1}{n} \cdot \frac{3^n}{3^{n+1}} = 1 \cdot \frac{1}{3} = \frac{1}{3}$. Since $L = 1/3 < 1$, the series converges.

9 What is the limit L in the Ratio Test for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

a. 1 **(test is inconclusive)** ✓

b. 0

c. ∞

d. $1/2$

$\lim_{n\to\infty} \frac{1/(n+1)}{1/n} = \lim \frac{n}{n+1} = 1$. The Ratio Test is inconclusive ($L = 1$); the harmonic series diverges by the p-series test ($p = 1$).

Section 4: Radius and Interval of Convergence

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10 Find the radius of convergence R for the power series $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$.

a. $R = \frac{1}{2}$ ✓

b. $R = 2$

c. $R = 1$

d. $R = \infty$

Ratio test: $\left|\frac{(2x)^{n+1}}{n+1} \cdot \frac{n}{(2x)^n}\right| = 2|x|\frac{n}{n+1} \to 2|x|$. For convergence: $2|x| < 1 \Rightarrow |x| < 1/2$. Thus $R = 1/2$.

11 What is the radius of convergence of the Maclaurin series for e^x?

a. $R = \infty$ ✓

b. $R = 1$

c. $R = e$

d. $R = 0$

The exponential series converges for all complex numbers $x \in \mathbb{C}$; its radius of convergence is infinite ($R = \infty$).

12 For $\sum_{n=0}^{\infty} x^n$, what is the exact interval of convergence?

a. $(-1, 1)$ ✓

b. $[-1, 1]$

c. $(-1, 1]$

d. $[-1, 1)$

Geometric series with ratio x: converges if and only if $|x| < 1$. At endpoints $x = \pm 1$, terms do not approach 0, so neither endpoint converges.

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